

(Continuing from the previous section...)

...is excessively encrypted for concealment purposes (Obfuscation), the file is blocked at the gateway before it is ever executed and before it can acquire Ring 0 privileges.

6. Autonomous Execution and Instant Memory Freezing (Quarantine)

- Upon detecting an anomaly in a monitored program, the "Chef" eschews risky manipulations such as "Fake Shutdowns."
- Leveraging the capabilities provided by its physical isolation (TEE), it transmits a direct "Interrupt" signal to the main processor, instantaneously freezing the RAM block where the suspect program is operating (Memory Freeze).
- It subsequently duplicates the malicious software into its isolated Black Box and immediately eradicates any remnants from the host system.

Development and Pivot Roadmap

To operationalize this architecturally completed manifesto, the following phases will proceed in synchronization with the university academic calendar:

- **Phase 1: Proof of Concept (Construction of the Customs Officer)**
 - During the transition into the intensive algorithms and Data Structures coursework of the freshman year, the initial ML (Machine Learning) model will be trained utilizing Python and C++ directly, circumventing the complexities of hardware architecture.
 - This lightweight system, which quantifies file entropy and detects encrypted viruses, will serve as the project's first tangible deliverable.
- **Phase 2: Kernel Penetration (eBPF and Linux)**
 - Once Operating Systems concepts are fully consolidated, the development environment will migrate from the closed ecosystem of Windows to the open-source Linux kernel.
 - Utilizing eBPF technology, the "System Calls" of programs will be intercepted and monitored directly from the core of the operating system (Ring 0).
- **Phase 3: Hardware Integration (The Genesis of the Chef)**
 - Upon perfecting the foundational software architecture, the codebase will be directly integrated into the NPU (Neural Processing Unit) hardware of the 8945HX processor.
 - This will achieve complete physical isolation from the main system, thereby actualizing the "Trusted Execution Environment" (TEE).